

This is never possible in off-grid plants, as self-use power needs are widely variable and rated capacity is kept high to primarily meet peak demand. Thus all such plants have a large portion of their peak capacity available to feed into the grid and thereby harvest full energy.

Only when all such excess power is injected in a grid that one can harvest more solar power from investments made by the country under the scheme.

I am sure huge private investments will be made only when an identical subsidised rate per unit is paid to smaller grid-fed solar plants for excess power that such plants will have.

This is how it happens in advanced nations like the US, Germany and Spain, and even in China. Small non-industrial towns usually require power at night. So, say, a 50kW plant with grid-fed capability can deliver its power into the local high-impedance grid during the day and draw that power at night. This will then sharply reduce the need for storing all power in batteries.

## Grid points

Studies in Spain reveal that feeding even 100 kW of solar power in a low-voltage grid immediately improves the power delivered to local people. So for bringing about quick improvement in availability of quality power in energy-starved areas, our best option is to follow the Europeans and Americans and feed solar power into a local low-voltage grid. The technology of measuring such energy feed is universally well-established and used everywhere else in the world, enabling the power utility to pay those who feed solar power into the grid.

There are many more reasons to promote distributed solar power generation than buying solar power only from big plants of one megawatt and above.

The first and the foremost reason is the fact that there is practically no financial advantage in having just large plants. The investment in a solar

power plant is the same per watt of solar power. A 10kW plant will cost just 10 per cent of the cost of 100kW plant or the cost of a 1MW plant will be ten times that of a 100kW plant. This is because the basic PV panel is the most important part of the plant cost and its power capacity is 250Wp to 300Wp.

One just has to add more panels to build a larger plant. Since the economy of scale doesn't give any advantage, distributed generation is considered to be a better concept than generating in bulk. This, in turn, means that such smaller but more effective solar energy plants could be created by medium-size enterprises for local areas, creating local jobs.

The second reason relates to the location of the feed into the copper grid. Feeding power at high-impedance nodes will cause immediate power-quality improvement. Consider ten 10-50kW solar plants in areas near a town feeding their power into the local lower-voltage grid. This will immediately improve the stability and reliability of the power available to the local people served from that feeder. In such a case, many local investors will come forward to invest in several smaller power plants within a town or a village. Investment in all these plants will also be profitable if the government buys power from these plants at the same rate as they currently pay to big plants feeding power into the 33kV grid.

## Benefits of change

Feeding surplus power into the grid from 'self-use' solar generators will also improve the power quality in the neighbourhood. If an industry in a small town sets up, say, a 50 or 100kW plant for its own captive use, it always has some excess power to feed into the grid. If this excess solar power is fed into the grid, it will not only help the country harvest more power but also load such plants to their full capacity so long as the sun is shining. Once this happens, a large number of local entrepreneurs will set up such small plants. Experiences of solar power installations in the world also show that

smaller solar power plants are easier to maintain than bigger plants. This also helps in facilitating local employment.

One major problem with solar power plants is the unusually large ground space that they need per kilowatt of power. As much as one square metre of area is needed for generating 100W of peak power! Location has to be such that there are no trees and structures casting shadows on the solar array. The place also needs to be secure and free from distractions like birds and monkeys. It is easier to find such a place on terraces and barren non-arable lands in towns and fringe rural areas, each accommodating 10 to 100kW plants. SMEs can therefore get into power generation business. The ministry of SME should seriously use this opportunity to ensure that SMEs benefit from this as much as big corporate enterprises.

## Is it unfair policy for small enterprises?

Finally, the government should not forget that one can challenge the current policy of buying power only from big solar plants in which only large corporations can invest money.

Suppose in a remote village, local people invest and develop power with a 20kW plant to provide power to 50 to 100 hamlets. How can they be denied subsidies at the same rate as for a big corporation producing over one megawatt of power.

No court will allow the government to discriminate in such a high-handed manner. In fact, such villages that invest in their own power plants need to be encouraged, as the state has been unable to invest and feed power to that village. Before the matter reaches that legal tangle, it is better that the government looks at its subsidy policy and ensures that there is justice for all! ●

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